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Speech Recognition Using Active Acoustic Sensing

Abstract

Techniques and apparatuses are described that perform speech recognition using active acoustic sensing. During active acoustic sensing, a hearable transmits and receives at least one ultrasound signal, which propagates within a user's ear canal. This ultrasound signal can be modulated by a user's speech as well as by other muscle movements associated with speech (e.g., jaw movement and/or tongue movement). As such, the ultrasound signal contains information that is correlated with speech as well as additional contextual information in how the user created the speech using their body. With active acoustic sensing, the hearable can directly perform speech recognition based on the ultrasound signal and/or enhance speech recognition by fusing information derived from the ultrasound signal with information derived from an audible signal that is passively sensed using a microphone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/559,576, filed on Feb. 29, 2024, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Wireless technology has become prevalent in everyday life, making communication and data readily accessible to users. One type of wireless technology are wireless hearables, examples of which include wireless earbuds and wireless headphones. Wireless hearables have allowed users freedom of movement while listening to audio content from music, audio books, podcasts, and videos. With the prevalence of wireless hearables, there is a market for adding additional features to existing hearables without introducing hardware changes.

SUMMARY

[0003] Techniques and apparatuses are described for performing speech recognition using active acoustic sensing. During active acoustic sensing, a hearable transmits and receives at least one ultrasound signal, which propagates within a user's ear canal. This ultrasound signal can be modulated by a user's speech as well as by other muscle movements associated with speech (e.g., jaw movement). As such, the ultrasound signal contains information that is correlated with speech as well as additional contextual information in how the user created the speech using their body. With active acoustic sensing, the hearable can directly perform speech recognition based on the ultrasound signal and/or enhance speech recognition by fusing information derived from the ultrasound signal with information derived from an audible signal that is passively sensed using a microphone.

[0004] Aspects described below include a method for performing speech recognition using active acoustic sensing. The method includes transmitting, during a first time period, an ultrasound transmit signal that propagates within at least a portion of an ear canal of a user. The method also includes receiving, during the first time period, an ultrasound receive signal. The ultrasound receive signal represents a version of the ultrasound transmit signal with one or more characteristics modified based on the propagation within the ear canal and based on the user speaking a phrase during at least a portion of the first time period. The method additionally includes recognizing the spoken phrase based on the ultrasound receive signal. The method can optionally include generating a control signal that controls an operation of a device based on the recognized spoken phrase or converting the recognized spoken phrase to text.

[0005] Aspects described below include a computer-readable storage medium comprising instructions that, responsive to execution by a processor, cause a hearable to perform any one of the methods described herein.

[0006] Aspects described below include a device with at least one transducer and at least one processor. The device is configured to perform, using the at least one transducer and the at least one processor, any one of the methods described herein.

[0007] Aspects described below include a system with means for performing speech recognition using active acoustic sensing.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] Apparatuses for and techniques that perform speech recognition using active acoustic sensing are described with reference to the following drawings. The same numbers are used throughout the drawings to reference like features and components:

[0009] FIG. 1 illustrates an example environment in which speech recognition using active acoustic sensing can be implemented;

[0010] FIG. 2 illustrates an example environment in which speech recognition using active acoustic sensing can be implemented;

[0011] FIG. 3 illustrates example components of a computing device;

[0012] FIG. 4 illustrates example components of a hearable;

[0013] FIG. 5 illustrates example operations of two hearables;

[0014] FIG. 6 illustrates an example implementation of a hearable capable of performing speech recognition using active acoustic sensing;

[0015] FIG. 7 illustrates example components of a speech-recognition module for performing aspects of speech recognition using active acoustic sensing;

[0016] FIG. 8 illustrates a first example implementation of a speech-recognition module;

[0017] FIG. 9 illustrates a second example implementation of a speech-recognition module;

[0018] FIG. 10 illustrates an impact of a first spoken phrase on an ultrasound receive signal;

[0019] FIG. 11 illustrates an impact of a second spoken phrase on an ultrasound receive signal;

[0020] FIG. 12 illustrates an impact of a third spoken phrase on an ultrasound receive signal;

[0021] FIG. 13 illustrates an example method for performing speech recognition using active acoustic sensing;

[0022] FIG. 14 illustrates another example method for performing speech recognition using active acoustic sensing; and

[0023] FIG. 15 illustrates an example computing system embodying, or in which techniques may be implemented that enable use of, speech recognition using active acoustic sensing.

DETAILED DESCRIPTION

[0024] As electronic devices become more ubiquitous, users incorporate them into everyday life. A user, for example, may use an electronic device to get daily weather and traffic information, control a temperature of a home, answer a doorbell, turn on or off a light, and/or play background music. Interacting with some electronic devices, however, can be cumbersome and inefficient. An electronic device, for instance, can have a physical user interface that may require a user to navigate through one or more prompts by physically touching the electronic device. In this case, the user has to devote attention away from other primary tasks to interact with the electronic device, which can be inconvenient and disruptive.

[0025] To address this problem, some electronic devices support voice control, which enables a user to interact with the electronic device in a non-physical and less cognitively demanding way compared to other interfaces that require physical touch and/or the user's visual attention. With voice control, the electronic device seamlessly exists in the surrounding environment and provides the user access to information and services while the user performs a primary task, such as cooking, cleaning, driving, talking with people, or reading a book. For voice control, the electronic device detects a user's speech and recognizes a phrase (or command) that is spoken by the user.

[0026] While voice control can provide a convenient means of interacting with an electronic device, there are several challenges associated with performing speech recognition. In a noisy environment, for instance, the user's voice can be made imperceptible by the other external noise. Consequently, it can be challenging to detect and/or recognize phrases spoken by the user. Also, sometimes the noisy environment can cause the voice control to incorrectly respond to a voice of another person who is not authorized to use the electronic device. Although it may be easier to perform speech recognition in a quiet environment, the quiet environment can pose other

challenges. In a library or a classroom, for instance, it may be inappropriate and/or socially awkward to speak audibly. As such, a user may forego the use of voice control in these types of situations.

[0027] To improve aesthetics and reduce encumbrance, it can be desirable to design hearables with smaller sizes. As space becomes limited, it can be challenging to integrate additional components, such as the voice accelerometer, within the hearables. With the prevalence of hearables, there is a market for adding additional features to existing hearables to facilitate speech recognition without introducing hardware changes.

[0028] Provided according to one or more preferred embodiments is a hearable, such as an earbud, that is capable of performing a novel physiological monitoring process termed herein audioplethysmography. Audioplethysmography is an active acoustic method capable of sensing subtle physiologically-related changes observable at a user's outer and middle ear. Instead of relying on other auxiliary sensors, such as optical or electrical sensors, audioplethysmography involves transmitting and receiving ultrasound signals that at least partially propagate within a user's ear canal. To perform audioplethysmography, the hearable forms at least a partial seal in or around the user's outer ear. This seal enables formation of an acoustic circuit, which includes the seal, the hearable, the ear canal, and an ear drum of the ear. By transmitting and receiving ultrasound signals, the hearable can recognize changes in the acoustic circuit to perform speech recognition. Speech recognition involves identifying a phrase that is spoken by the user. The phrase can include any sound that is produced using the user's lung's, vocal cords, and/or mouth. Example types of vocalizations can involve the user speaking, whispering, shouting, humming, whistling, singing, or making other utterances.

[0029] During active acoustic sensing, a hearable transmits and receives at least one ultrasound signal, which propagates within the user's ear canal. This ultrasound signal can be modulated by a user's speech as well as by other muscle movements associated with speech (e.g., jaw movement). As such, the ultrasound signal contains information that is correlated with speech as well as additional contextual information in how the user created the speech using their body. With active acoustic sensing, the hearable can directly perform speech recognition based on the ultrasound signal and/or enhance speech recognition by fusing information derived from the ultrasound signal with information derived from an audible signal that is passively sensed using a microphone.

[0030] Utilizing active acoustic sensing for speech recognition can provide several benefits. In a first aspect, active acoustic sensing enables the hearable to support a discrete version of voice control that involves silent speech. With silent speech, the user can perform the muscle movements associated with speaking without producing an audible sound. In this case, speech recognition using active acoustic sensing provides a discreet and socially acceptable means of controlling a device in various environments. This feature can be particularly advantageous in environments, such as a library or a classroom, where it may be inappropriate and/or socially awkward to speak audibly.

[0031] In a second aspect, active acoustic sensing can further improve speech recognition in a noisy and/or loud environment. This is because the ultrasound signal associated with active acoustic sensing is less susceptible to noise that is present in an external environment compared to an over-the-air audible signal. In a third aspect, active acoustic sensing can provide additional contextual information that can improve speech recognition for users who have difficulty speaking, including those having a motor speech disorder (e.g., dysarthria). In addition to being relatively unobtrusive, some hearables can be configured to support speech recognition using active acoustic sensing without the need for additional hardware. As such, the size, cost, and power usage of the hearable can help make speech recognition accessible to a larger group of people and improve the user experience with hearables.

[0032] Active acoustic sensing can improve the performance of speech recognition relative to other sensing techniques. Techniques that involve a non-wearable electronic device capable of observing

the user's jaw movement using ultrasound, for instance, may not be as sensitive or as accurate compared to active acoustic sensing. This is in part because active acoustic sensing can be performed using a hearable that is worn by the user, which enables the user's vocalization to be directly measured based on a pressure wave that propagates to the user's ear. In contrast, observing the user's jaw movement with ultrasound may only work in limited circumstances in which the user is properly oriented relative to the electronic device and the electronic device has an unobstructed line-of-sight to the user's face to observe the jaw movement. Still other techniques utilize a microphone to detect voice through bone conduction. While this passive technique can provide some additional information associated with speech, the information is limited to the low frequency band, which can include frequencies below 1.5 kilohertz (kHz). In contrast, active acoustic sensing can provide feature-rich information associated with both the low frequency band (e.g., below 1.5 kHz) and a high frequency band (e.g., above 1.5 kHz).

Operating Environment

[0033] FIG. 1 is an illustration of an example environment **100** in which active acoustic sensing can be implemented. In the example environment **100**, a hearable **102** is connected to a computing device **104** using a physical or wireless interface. The hearable **102** is a device that can play audible content provided by the computing device **104** and direct the audible content into a user **106**'s ear **108**. In this example, the hearable **102** operates together with the computing device **104**. In other examples, the hearable **102** can operate or be implemented as a stand-alone device. Although depicted as a smartphone, the computing device **104** can include other types of devices, including those described with respect to FIG. 3.

[0034] The hearable **102** is capable of performing audioplethysmography **110**, which is an active acoustic method of sensing that occurs at the ear **108**. The hearable **102** can perform this sensing without the use of other auxiliary sensors, such as an optical sensor or an electrical sensor. Through audioplethysmography **110**, the hearable **102** can perform speech recognition **112**. Speech recognition **112** enables the hearable **102** (or the computing device **104**) to recognize one or more phrases that are spoken by the user **106**. A phrase can involve any type of vocalization associated with speaking, whispering, shouting, humming, whistling, singing, or other utterances. The phrase can include a single word or multiple words. Generally speaking, speech recognition **112** can also be referred to as automatic speech recognition (ASR), computer speech recognition, or speech-to-text (STT). Two types of speech recognition **112** are further described below. The techniques for performing speech recognition **112** through active acoustic sensing can generally be performed using a hearable **102** that is worn in or on an ear of any person.

[0035] A first type of speech recognition **112** is referred to as ultrasound-based speech recognition. Ultrasound-based speech recognition involves speech recognition **112** that can be directly performed using one or more signals derived from the ultrasound signals associated with audioplethysmography **110**. This type of speech recognition **112** can be performed without the use of passive audio sensing (e.g., without generating an audio signal that represents an audible over-the-air signal). Also, ultrasound-based speech recognition can be performed without relying on other types of sensors (e.g., sensors that do not utilize ultrasound), such as a voice accelerometer. With ultrasound-based speech recognition, the user **106** can optionally speak a phrase with minimal or no sound (e.g., by whispering and/or by performing silent speech). Other speech recognition techniques that rely on passive audio sensing may be unable to support silent speech as there is an absence of audible sound.

[0036] A second type of speech recognition **112** is referred to as ultrasound-fusion speech recognition. Ultrasound-fusion speech recognition involves speech recognition **112** that can be performed using a combination of information derived from audioplethysmography **110** and information derived from passive audio sensing. This fusion can enhance the performance of speech recognition **112** in noisy and/or loud environments relative to other techniques that do not utilize information derived from audioplethysmography **110**. It can also support the use of silent

speech. In some cases, ultrasound-fusion speech recognition can obviate the use of other sensors, such as a voice accelerometer. In addition to enabling the hearable **102** to be implemented without these other sensors, which can be bulky and/or expensive, audioplethysmography **110** can be less susceptible to noise caused by the user moving (e.g., walking or running). As such, audioplethysmography **110** can further enhance the performance of speech recognition while the user is moving relative to other types of sensors.

[0037] To perform speech recognition **112**, the hearable **102** uses audioplethysmography **110** to detect subtle pressure waves that propagate to the user **106**'s ear canal **114**. These pressure waves modify characteristics of ultrasound signals that are transmitted and received by the hearable **102** and propagate through the ear canal **114**. As the user **106** speaks, the ear canal **114** deforms at least in part due to the speech itself and at least in part due to the muscle movements associated with performing the speech. As such, at least a portion of the received ultrasound signal includes information that is correlated with the user **106**'s speech and at least another portion of the received ultrasound signal includes other information that is correlated with muscle movements associated with generating the speech. In some cases, the user **106**'s speech can be directly reconstructed from the received ultrasound signal.

[0038] To use audioplethysmography **110**, the user **106** positions the hearable **102** in a manner that creates at least a partial seal **116** around or in the ear **108**. Some parts of the ear **108** are shown in FIG. **1**, including the ear canal **114** and an ear drum **118** (or tympanic membrane). Due to the seal **116**, the hearable **102**, the ear canal **114**, and the ear drum **118** couple together to form an acoustic circuit. Audioplethysmography **110** involves, at least in part, measuring properties associated with this acoustic circuit. The properties of the acoustic circuit can change due to a variety of different situations or actions.

[0039] For example, consider a change that occurs in a physical structure of the ear **108**. Example changes to the physical structure include a change in a geometric shape of the ear canal **114** and/or a change in a volume of the ear canal **114**. This change can be caused, at least in part, by a pressure wave associated with the user **106**'s speech. For instance, the tissue around the ear canal **114** and the ear drum **118** itself are slightly “squeezed” due to the bone conduction and/or the pressure wave. This squeeze causes a volume of the ear canal **114** to be slightly reduced. As the squeezing subsides, the volume of the ear canal **114** is slightly increased. The increasing and decreasing of the volume of the ear canal **114** is indicated by the arrows in FIG. **1**. The physical changes within the ear **108** can modulate an amplitude and/or phase of an ultrasound signal that propagates through the ear canal **114**.

[0040] The techniques for audioplethysmography **110** can be performed while the hearable **102** is rendering (e.g., playing or transmitting) audible content and/or while the user **106** is actively moving or performing an activity. As such, active acoustic sensing enables the hearable **102** to perform speech recognition **112** in a variety of different situations. One such situation is further described with respect to FIG. **2**.

[0041] FIG. **2** illustrates an example environment **200** in which speech recognition **112** using active acoustic sensing can be implemented. The environment **200** represents a noisy and/or loud environment that includes a variety of audible signals. These audible signals propagate over-the-air and can make it challenging for the user **106** to utilize a voice user interface **202** of the computing device **104**. Example noise sources include environmental noise **204**, music **206** played by a speaker **208**, a vocalization **210** made by another person, or some combination thereof.

[0042] The noise sources can make it challenging for the voice user interface **202** to detect and/or recognize a phrase **212** that is spoken by the user **106**. In some cases, the phrase **212** can be a voiceprint phrase or a voice command. The voiceprint phrase can be a unique phrase that enables the user **106** to be identified and authenticated for voice-control access. The voice command is another unique phrase that can be used to control the hearable **102** and/or the computing device **104**.

[0043] To address this problem, the user **106** speaks the phrase **212** while wearing the hearable **102**. With speech recognition **112**, the hearable **102** can identify the phrase **212** and pass this information to the voice user interface **202** of the computing device **104**. Additionally or alternatively, speech recognition **112** can be used to support other applications of the computing device **104**, as further described with respect to FIG. 3.

[0044] FIG. 3 illustrates an example implementation of the computing device **104**. The computing device **104** is illustrated with various non-limiting example devices including a desktop computer **104-1**, a tablet **104-2**, a laptop **104-3**, a television **104-4**, a computing watch **104-5**, computing glasses **104-6**, a gaming system **104-7**, a microwave **104-8**, and a vehicle **104-9**. Other devices may also be used, such as an augmented and/or virtual reality headset, a home service device, a smart speaker, a smart thermostat, a baby monitor, a Wi-Fi™ router, a drone, a trackpad, a drawing pad, a netbook, an e-reader, a home automation and control system, a wall display, and another home appliance. Note that the computing device **104** can be wearable, non-wearable but mobile, or relatively immobile (e.g., desktops and appliances).

[0045] The computing device **104** includes one or more computer processors **302** and at least one computer-readable medium **304**, which includes memory media and storage media. Applications and/or an operating system (not shown) embodied as computer-readable instructions on the computer-readable medium **304** can be executed by the computer processor **302** to provide some of the functionalities described herein. The computer-readable medium **304** can optionally include an application **306**, the voice user interface **202**, and/or a voice authenticator **308**. Additionally or alternatively, the computer-readable medium **304** can include other types of applications that utilize speech recognition **112**, such as a translation service that converts speech to text.

[0046] The application **306** can use information provided by the hearable **102** to perform an action. Example actions can include displaying data associated with audioplethysmography **110** to the user **106**. For speech recognition **112**, the application **306** can indicate whether or not the phrase **212** is recognized. The voice user interface **202** can enable the user **106** to control the computing device **104** via voice commands, as described with respect to FIG. 2. The voice authenticator **308** can authenticate the user **106** and enable use of the voice user interface **202** upon successful authentication. The application **306**, the voice user interface **202**, and/or the voice authenticator **308** can utilize aspects of speech recognition **112** to provide certain features and/or enhance security of the computing device **104**.

[0047] The computing device **104** can also include a network interface **310** for communicating data over wired, wireless, or optical networks. For example, the network interface **310** may communicate data over a local-area-network (LAN), a wireless local-area-network (WLAN), a personal-area-network (PAN), a wide-area-network (WAN), an intranet, the Internet, a peer-to-peer network, point-to-point network, a mesh network, Bluetooth®, and the like. The computing device **104** may also include the display **312**. Although not explicitly shown, the hearable **102** can be integrated within the computing device **104**, or can connect physically or wirelessly to the computing device **104**. The hearable **102** is further described with respect to FIG. 4.

[0048] FIG. 4 illustrates an example hearable **102**. The hearable **102** is illustrated with various non-limiting example devices, including wireless earbuds **402-1**, wired earbuds **402-2**, and headphones **402-3**. The earbuds **402-1** and **402-2** are a type of in-ear device that fits into the ear canal **114**. Each earbud **402-1** or **402-2** can represent a hearable **102**. Headphones **402-3** can rest on top of or over the ears **108**. The headphones **402-3** can represent closed-back headphones, open-back headphones, on-ear headphones, or over-ear headphones. Each headphone **402-3** includes two hearables **102**, which are physically packaged together. In general, there is one hearable **102** for each ear **108**. The headphones **402-3** may be designed in some manner or may utilize techniques, such as beamforming, to assist with directing signals used for audioplethysmography **110** into the ear canal **114**.

[0049] The hearable **102** includes a communication interface **404** to communicate with the

computing device **104**, though this need not be used when the hearable **102** is integrated within the computing device **104**. The communication interface **404** can be a wired interface or a wireless interface, in which audio content is passed from the computing device **104** to the hearable **102**. The hearable **102** can also use the communication interface **404** to pass data associated with audioplethysmography **110** and/or speech recognition **112** to the computing device **104**. In general, the data provided by the communication interface **404** is in a format usable by the application **306**, the voice user interface **202**, the voice authenticator **308**, or another application of the computing device **104**.

[0050] The communication interface **404** also enables the hearable **102** to communicate with another hearable **102**. During bistatic sensing, for instance, the hearable **102** can use the communication interface **404** to coordinate with the other hearable **102** to support two-ear audioplethysmography **110**, as further described with respect to FIG. 5. In particular, the transmitting hearable **102** can communicate timing and waveform information to the receiving hearable **102** to enable the receiving hearable **102** to appropriately demodulate a received ultrasound signal.

[0051] The hearable **102** includes at least one transducer **406** that can convert electrical signals into sound waves. The transducer **406** can also detect and convert sound waves into electrical signals. These sound waves may include ultrasonic frequencies, which may be used for audioplethysmography **110**. In particular, a frequency spectrum (e.g., range of frequencies) that the transducer **406** uses to generate an ultrasound signal can include frequencies from the ultrasonic range, e.g., between 20 kHz to 2 megahertz (MHz). Other example frequency spectrums for audioplethysmography **110** can encompass frequencies between 20 and 60 kHz or between 30 and 40 kHz.

[0052] In an example implementation, the transducer **406** has a monostatic topology. With this topology, the transducer **406** can convert the electrical signals into sound waves and convert sound waves into electrical signals (e.g., can transmit or receive acoustic and/or ultrasound signals). Example monostatic transducers may include piezoelectric transducers, capacitive transducers, and micro-machined ultrasonic transducers (MUTs) that use microelectromechanical systems (MEMS) technology.

[0053] Alternatively, the transducer **406** can be implemented with a bistatic topology, which includes multiple transducers that are physically separate. In this case, a first transducer converts the electrical signal into sound waves (e.g., transmits acoustic and/or ultrasound signals), and a second transducer converts sound waves into an electrical signal (e.g., receives the acoustic and/or ultrasound signals). An example bistatic topology can be implemented using at least one speaker **408** and at least one microphone **410**. The speaker **408** and the microphone **410** can be dedicated for audioplethysmography **110** or can be used for both audioplethysmography **110** and other functions of the computing device **104** (e.g., passive audio sensing, presenting audible content to the user **106**, capturing the user **106**'s voice for a phone call, or for voice control).

[0054] In general, the speaker **408** and the microphone **410** are directed towards the ear canal **114** (e.g., oriented towards the ear canal **114**). Accordingly, the speaker **408** can direct ultrasound signals towards the ear canal **114**, and the microphone **410** is responsive to receiving ultrasound signals from the direction associated with the ear canal **114**. In some cases, the hearable **102** includes another microphone **410** that is directed away from the ear canal **114** towards an external environment (e.g., oriented away from the ear canal **114**). This other microphone can be used to receive over-the-air signals, which can include the user **106**'s voice and/or environmental noise.

[0055] The hearable **102** includes at least one analog circuit **412**, which includes circuitry and logic for conditioning electrical signals in an analog domain. The analog circuit **412** can include analog-to-digital converters, digital-to-analog converters, amplifiers, filters, mixers, and switches for generating and modifying electrical signals. In some implementations, the analog circuit **412** includes other hardware circuitry associated with the speaker **408** or microphone **410**.

[0056] The hearable **102** also includes at least one system processor **414** and at least one system medium **416** (e.g., one or more computer-readable storage media). In the depicted configuration, the system medium **416** includes a pre-processing module **418** and a speech-recognition module **420**. The system medium **416** also optionally includes a calibration module **422**. The pre-processing module **418**, the speech-recognition module **420**, and the calibration module **422** can be implemented using hardware, software, firmware, or a combination thereof. In this example, the system processor **414** implements the pre-processing module **418**, the speech-recognition module **420**, and the calibration module **422**. In an alternative example, the computer processor **302** of the computing device **104** can implement at least a portion of the pre-processing module **418**, the speech-recognition module **420**, and/or the calibration module **422**. In this case, the hearable **102** can communicate digital samples of the acoustic and/or ultrasound signals to the computing device **104** using the communication interface **404**.

[0057] Operations of the pre-processing module **418**, the speech-recognition module **420**, and the calibration module **422** are further described with respect to FIG. 6. Aspects of speech recognition **112** using active acoustic sensing can be performed, at least partially, by the speech-recognition module **420**, as further described with respect to FIG. 7.

[0058] Some hearables **102** include an active-noise-cancellation circuit **424**, which enables the hearables **102** to reduce background or environmental noise. In this case, the microphone **410** used for audioplethysmography **110** can be implemented using a feedback microphone of the active-noise-cancellation circuit **424**. During active noise cancellation, the feedback microphone provides feedback information regarding the performance of the active noise cancellation. During audioplethysmography **110**, the feedback microphone receives an ultrasound signal, which is provided to the pre-processing module **418**. In some situations, active noise cancellation and audioplethysmography **110** are performed simultaneously using the feedback microphone. In this case, the ultrasound signal received by the feedback microphone can be provided to the pre-processing module **418** and the feedback signal for active noise cancellation can be provided to the active-noise-cancellation circuit **424**. Other implementations are also possible in which the microphone **410** is implemented using a feedforward microphone of the active-noise-cancellation circuit **424**. In some implementations, the feedforward microphone performs passive audio sensing to provide an audio signal for denoising operations and/or for aspects of ultrasound-fusion speech recognition.

[0059] Although not explicitly shown in FIG. 4, the system medium **416** can also include a voice user interface **202**, a voice authenticator **308**, and/or another application that utilizes speech recognition **112**. In this case, the voice user interface **202** enables the user **106** to use voice controls to control an operation of the hearable **102**. The voice authenticator **308** can authenticate the user **106** and enable the voice user interface **202** for the hearable **102**. Different types of audioplethysmography **110** are further described with respect to FIG. 5.

Active Acoustic Sensing

[0060] FIG. 5 illustrates example operations of two hearables **102-1** and **102-2**. In a first example operation, the hearables **102-1** and **102-2** perform single-ear audioplethysmography **110**. This means that the hearables **102-1** and **102-2** independently perform audioplethysmography **110** on different ears **108** of the user **106**. In this case, the first hearable **102-1** is proximate to the user **106**'s right ear **108**, and the second hearable **102-2** is proximate to the user **106**'s left ear **108**. Each hearable **102-1** and **102-2** includes a speaker **408** and a microphone **410**. The hearables **102-1** and **102-2** can operate in a monostatic manner during the same time period or during different time periods. In other words, each hearable **102-1** and **102-2** can independently transmit and receive ultrasound signals.

[0061] For example, the first hearable **102-1** uses the speaker **408** to transmit a first ultrasound transmit **502-1**, which propagates within at least a portion of the user **106**'s right ear canal **114**. The first hearable **102-1** uses the microphone **410** to receive a first ultrasound receive signal **504-1**. The

first ultrasound receive signal **504-1** represents a version of the first ultrasound transmit signal **502-1** that is modified, at least in part, by the acoustic circuit associated with the right ear canal **114**. This modification can change an amplitude, phase, and/or frequency of the first ultrasound receive signal **504-1** relative to the first ultrasound transmit signal **502-1**.

[0062] Similarly, the second hearable **102-2** uses the speaker **408** to transmit a second ultrasound transmit signal **502-2**, which propagates within at least a portion of the user **106**'s left ear canal **114**. The second hearable **102-2** uses the microphone **410** to receive a second ultrasound receive signal **504-2**. The second ultrasound receive signal **504-2** represents a version of the second ultrasound transmit signal **502-2** that is modified by the acoustic circuit associated with the left ear canal **114**. This modification can change an amplitude, phase, and/or frequency of the second ultrasound receive signal **504-2** relative to the second ultrasound transmit signal **502-2**.

[0063] The techniques of single-ear audioplethysmography **110** can be particularly beneficial as it enables the computing device **104** to compile information from both hearables **102-1** and **102-2**, which can further improve measurement confidence. For some aspects of audioplethysmography **110**, it can be beneficial to analyze the acoustic channel between two ears **108**, as further described below.

[0064] In a second example operation, the two hearables **102-1** and **102-2** perform two-ear audioplethysmography **110**. This means that the hearables **102-1** and **102-2** jointly perform audioplethysmography **110** across two ears **108** of the user **106**. In this case, at least one of the hearables **102** (e.g., the first hearable **102-1**) includes the speaker **408**, and at least one of the other hearables **102** (e.g., the second hearable **102-2**) includes the microphone **410**. The hearables **102-1** and **102-2** operate together in a bistatic manner during the same time period.

[0065] During operation, the first hearable **102-1** transmits a third ultrasound transmit **502-3** using the speaker **408**. The third ultrasound transmit signal **502-3** propagates through the user **106**'s right ear canal **114**. The third ultrasound transmit signal **502-3** also propagates through an acoustic channel that exists between the right and left ears **108**. In the left ear **108**, the third ultrasound transmit signal **502-3** propagates through the user **106**'s left ear canal **114** and is represented as a third ultrasound receive signal **504-3**. The second hearable **102-2** receives the third ultrasound receive signal **504-3** using the microphone **410**. The third ultrasound receive signal **504-3** represents a version of the third ultrasound transmit signal **502-3** that is modified by the acoustic circuit associated with the right ear canal **114**, modified by the acoustic channel associated with the user **106**'s face, and modified by the acoustic circuit associated with the left ear canal **114**. This modification can change an amplitude, phase, and/or frequency of the third ultrasound receive signal **504-3** relative to the third ultrasound transmit signal **502-3**. In some cases, the hearable **102-2** measures the time-of-flight (ToF) associated with the propagation from the first hearable **102-1** to the second hearable **102-2**. Sometimes a combination of single-ear and two-ear audioplethysmography **110** are applied to further improve measurement confidence.

[0066] The ultrasound transmit signals **502** of FIG. 5 can represent a variety of different types of signals as described above with respect to FIG. 4. In example implementations, the ultrasound transmit signal **502** can be a continuous-wave signal (e.g., a sinusoidal signal) or a pulsed signal. Some ultrasound transmit signals **502** can have a particular tone (or frequency). Other ultrasound transmit signals **502** can have multiple tones (or multiple frequencies). A variety of modulations can be applied to generate the ultrasound transmit signal **502**. Example modulations include linear frequency modulations, triangular frequency modulations, stepped frequency modulations, phase modulations, or amplitude modulations. The ultrasound transmit signal **502** can be transmitted as part of a calibration procedure or a measurement procedure, as further described as part of FIG. 6.

[0067] FIG. 6 illustrates an example implementation of the hearable **102** for performing speech recognition **112**. In the depicted configuration, the hearable **102** includes the speaker **408**, the microphone **410**, the analog circuit **412**, the pre-processing module **418**, the speech-recognition module **420**, and the calibration module **422**. Other implementations of the hearable **102**, however,

are also possible in which the hearable **102** does not include the calibration module **422** to reduce processing power requirements. In this case, the pre-processing module **418** can perform aspects of frequency selection as further described below to improve the signal-to-noise ratio for audioplethysmography **110**.

[0068] Outputs of the speaker **408** and the microphone **410** are coupled to inputs of the analog circuit **412**. The pre-processing module **418** has inputs that are coupled to outputs of the analog circuit **412**. The pre-processing module **418** also has an output that is coupled to inputs of the speech-recognition module **420** and the calibration module **422**. In an example implementation, the pre-processing module **418** includes at least one in-phase and quadrature mixer (I/Q mixer) and at least one filter. The in-phase and quadrature mixer performs frequency down-conversion and can be implemented using at least two mixers, at least one phase shifter, and at least one combiner (e.g., a summation circuit). The filter attenuates intermodulation products that are generated by the in-phase and quadrature mixer. In an example implementation, the filter is implemented using a low-pass filter.

[0069] The pre-processing module **418** can optionally include at least one frequency selector. The frequency selector can identify and select one or more tones (or carrier frequencies) that provide a high-quality signal for later processing. The frequency selector can further pass the selected tones to other processing modules (e.g., the speech-recognition module **420**) and filter (or attenuate) other tones that are not selected. The frequency selector can be implemented in a similar manner as the calibration module **422**, which is further described below.

[0070] The speech-recognition module **420** can optionally have another input that is coupled to the microphone **410** (or another microphone not shown). Example implementations of the speech-recognition module **420** are further described with respect to FIGS. 7-9. With the speech-recognition module **420**, the hearable **102** performs a measurement procedure that includes performing speech recognition **112** using audioplethysmography **110** (e.g., performing ultrasound-based speech recognition and/or ultrasound-fusion speech recognition).

[0071] The calibration module **422** has an output that is coupled to the speaker **408**. The calibration module **422** includes at least one frequency selector. The frequency selector can include at least one amplitude detector, at least one phase detector, at least one quality detector, and at least one comparator. Using the frequency selector, the calibration module **422** can perform a calibration procedure that determines appropriate characteristics (e.g., waveform or signal characteristics) of ultrasound transmit signals **502** to improve audioplethysmography **110** (e.g., to enhance the performance of speech recognition **112**). The calibration procedure enables audioplethysmography **110** to take into account the wear of the hearable **102** (e.g., the position of the hearable **102** relative to the ear canal **114**) and the physical structure of the ear canal **114** to determine a transmission frequency that can increase sensitivity.

[0072] Consider an example operation of the hearable **102** in accordance with single-ear audioplethysmography **110**. In this example, the hearable **102** includes the calibration module **422**. With the calibration module **422**, the hearable **102** can perform the calibration procedure prior to performing a measurement procedure. In some circumstances, the hearable **102** can perform on-head detection (or in-ear detection) by detecting the presence of the seal **116** and initiating the calibration procedure and/or the measurement procedure based on a determination that on-head detection is “true.” In other circumstances, the hearable **102** can initiate the calibration procedure based on a specified schedule or a timer, which can be controlled by the user **106** via the computing device **104**. The calibration procedure and the measurement procedure are further described below.

[0073] During both the calibration procedure and the measurement procedure, the speaker **408** transmits the ultrasound transmit signal **502** and the microphone **410** receives the ultrasound receive signal **504**. During the calibration procedure, the ultrasound transmit signal **502** and the ultrasound receive signal **504** can have tones **602-1** to **602-M**, where M represents a positive integer. The multiple tones **602-1** to **602-M** can be transmitted in parallel or in series over a given

time interval. In this case, the ultrasound transmit signal **502** can have a particular bandwidth on the order of several kilohertz. For example, the ultrasound transmit signal **502** can have a bandwidth of approximately 4, 5, 6, 8, 10, 16, or 20 kHz. In example implementations, the ultrasound transmit signal **502** is transmitted over multiple seconds, such as 2, 3, 4, 6, or more seconds. A duration of each tone **602** can be evenly divided over a total duration of the ultrasound transmit signal **502**. [0074] In an example implementation, the ultrasound transmit signal **502** for the calibration procedure can have seven tones **602** (e.g., M equals 7). In some cases, the tones **602** are evenly distributed across an interval. For example, the tones **602** can be in 1 kHz increments between 32 kHz and 38 kHz (e.g., at approximately 32, 33, 34, 35, 36, 37, and 38 kHz). The term “approximately” means that the tones **602** can be within 5% of a given value or less (e.g., within 3%, 2%, or 1% of the given value).

[0075] An amplitude of the calibration procedure's ultrasound transmit signal **502** can be approximately the same across the tones **602-1** to **602-M**. In this manner, power is evenly distributed across each tone **602**. The quantity of tones **602** (e.g., M) can be determined based on an output power of the speaker **408**. Increasing the quantity of tones **602** can increase a likelihood that the hearable **102** can support speech recognition **112** across various conditions including user wear and a physical structure of the user **106**'s ear canal **114**. However, an amplitude of the ultrasound transmit signal **502** can be limited across these tones **602** based on the output power of the speaker **408**. Thus, the quantity of tones **602** can be optimized based on an amount of output power that is available for audioplethysmography **110**.

[0076] During the measurement procedure, the ultrasound transmit signal **502** and the ultrasound receive signal **504** can have selected tones **604-1** to **604-N**, where N represents a positive integer that is less than or equal to M. The selected tones **604-1** to **604-N** can represent a subset (sometimes a proper subset) of the tones **602-1** to **602-M**. The selected tones **604** can be transmitted in parallel or in series over a given time interval.

[0077] An amplitude of the measurement procedure's ultrasound transmit signal **502** can be approximately the same across the selected tones **604-1** to **604-N**. In this manner, power is evenly distributed across each selected tone. The amplitude of the measurement procedure's ultrasound transmit signal **502** can be higher than the amplitude of the calibration procedure's ultrasound transmit signal **502** because the available output power is distributed across fewer tones. Additionally or alternatively, a duration of each of the selected tones **604** of the measurement procedure's ultrasound transmit signal **502** can be longer than the duration of the tones **602** of the calibration procedure's ultrasound transmit signal **502**. The higher amplitude and/or the longer duration can further improve the signal-to-noise ratio performance of the hearable **102** for audioplethysmography **110**. By using a few selected tones **604** that were determined to improve signal-to-noise ratio performance, the measurement procedure can achieve a higher level of accuracy and sensitivity for speech recognition **112**.

[0078] The analog circuit **412** performs analog-to-digital conversion to generate a digital transmit signal **606** and a digital receive signal **608** based on the ultrasound transmit signal **502** and the ultrasound receive signal **504**, respectively. The pre-processing module **418** performs frequency downconversion and demodulation to generate at least one pre-processed signal **610** based on the digital transmit signal **606** and the digital receive signal **608**. The pre-processing module **418** can also apply filtering to generate the pre-processed signal **610**.

[0079] Optionally, as part of the calibration procedure, the calibration module **422** processes the pre-processed signal **610** to determine the selected tones **604-1** to **604-N**. The selected tones **604-1** to **604-N** can improve performance of audioplethysmography **110** during the measurement procedure. To determine the selected tones **604-1** to **604-N**, the calibration module **422** extracts the amplitude and/or phase of the pre-processed signal **610** using the amplitude detector and the phase detector, respectively. The quality detector of the calibration module **422** measures quality metrics for each tone (or frequency) of the pre-processed signal **610** and for each of the characteristics

(e.g., amplitude and/or phase). Example quality metrics can include peak-to-average ratios and/or signal-to-noise ratios. The peak-to-average ratio represents a peak intensity within a frequency range of interest divided by an average intensity within this frequency range. A higher quality metric indicates a higher-quality signal, or more generally, better performance for audioplethysmography **110**.

[0080] The comparator of the calibration module **422** can evaluate the quality metrics with respect to a threshold. In an example implementation, the comparator determines the selected tones **604-1** to **604-N** for a subsequent measurement procedure based on the frequencies associated with the quality metrics that are greater than or equal to a threshold. Additionally or alternatively, the comparator can evaluate the quality metrics with respect to each other. In an example implementation, the comparator determines one of the selected tones based on a frequency with the highest quality metric across the amplitude. Also, the comparator can determine one of the selected tones **604-1** to **604-N** based on a frequency with the highest quality metric across the phase. In other implementations, the comparator can determine a single selected tone based on a frequency having the highest quality metric associated with either the amplitude or the phase.

[0081] In general, the calibration module **422** enables the selected tones **604-1** to **604-N** to be dynamically adjusted prior to the measurement procedure based on a current environment, which can account for a wear of the hearable **102** (e.g., a current insertion depth and/or rotation), a physical structure of the user **106**'s ear canal **114**, and a response characteristic of the hearable **102** (e.g., speaker, microphone, and/or housing). In this manner, the calibration module **422** can improve the signal-to-noise ratio performance of the hearable **102** for the measurement procedure. The calibration module **422** can also determine which tones **604** generate ultrasound receive signals **504** with desired characteristics for speech recognition **112**. In general, the calibration procedure can be performed whether or not the user **106** is speaking.

[0082] The calibration module **422** communicates the selected tones **604-1** to **604-N** to the speaker **408** using a control signal. The speaker **408** accepts the control signal that identifies the selected tones **604-1** to **604-N** and can transmit a subsequent ultrasound transmit signal **502** for speech recognition **112** using the selected tones **604-1** to **604-N**. With the calibration procedure, the hearable **102** can dynamically adjust the transmission frequency (e.g., one or more carrier frequencies) each time the seal **116** is formed (e.g., based on the wear of the hearable **102**) and based on the unique physical structure of the ear **108**. Through this calibration procedure, the hearables **102** on different ears **108** may operate with one or more different ultrasound frequencies.

[0083] As part of the measurement procedure, the speech-recognition module **420** can perform aspects of speech recognition **112** using the pre-processed signal **610** to generate a recognized phrase **612**. The recognized phrase **612** can include a representation of a phrase **212** that is spoken by the user **106** and is recognized by the speech-recognition module **420**. The recognized phrase **612** can be communicated to the computing device **104** (e.g., to the application **306**, the voice user interface **202**, and/or the voice authenticator **308**). Additionally or alternatively, the recognized phrase **612** can be used to control an operation of the hearable **102** and/or the computing device **104**.

[0084] In FIG. **6**, the calibration procedure and the measurement procedure are described as individual procedures that occur at different time intervals. In particular, the calibration procedure occurs before the measurement procedure. This enables the ultrasound transmit signal **502** for the measurement procedure to be transmitted with fewer tones than the ultrasound transmit signal **502** used for the calibration procedure, which can increase signal-to-noise ratio performance for audioplethysmography **110**. In some implementations, however, the hearable **102** can have sufficient output power to perform the measurement procedure with the multiple tones **602-1** to **602-M** using a single ultrasound transmit signal **502**. In this case, aspects of the calibration module **422** can be integrated within the pre-processing module **418** via a frequency selector. This frequency selector can effectively pass the selected tones **604-1** to **604-N** to the speech-recognition

module **420**.

[0085] In some implementations, the microphone **410** (or another microphone not shown) can perform passive audio sensing to detect an over-the-air audible signal **614** during the measurement process. The over-the-air audible signal **614** can include the user **106**'s speech as well as any noise that is present within the external environment. During passive audio sensing, the microphone **410** generates an audio receive signal **616**, which can include the audible speech of the user **106**. With the audio receive signal **616**, the speech-recognition module **420** can perform ultrasound-fusion speech recognition based on the pre-processed signal **610** and the audio receive signal **616**.

[0086] If the user **106** speaks (silently or audibly) during a time that the ultrasound receive signal **504** is received, the ultrasound receive signal **504** and any signal derived from it (e.g., the pre-processed signal **610**) can include information that is correlated with the speech. This information is referred to as a voice component and can be associated with frequencies above 100 Hz. In general, the voice component within the ultrasound receive signal **504** can be similar to, but orthogonal to, the voice component that is present within the audio receive signal **616**. This is because the voice component within the ultrasound receive signal **504** is caused by a different physical phenomenon involving the deformation of the ear canal **114**. In contrast, the voice component within the audio receive signal **616** is caused by the passage of air through the body, the shape of the user **106**'s mouth, the force of aspiration, or the movement of the tongue. With the voice component, the user **106**'s speech can be reconstructed solely using the pre-processed signal **610**. Alternatively, the orthogonality of the voice component enables speech recognition **112** to be enhanced by using a combination of the pre-processed signal **610** and the audio receive signal **616**.

[0087] The ultrasound receive signal **504** can also include other contextual information that can enhance speech recognition **112** compared to techniques that rely solely on passive audio sensing. This information is referred to as a contextual component and can be associated with frequencies below 100 Hz (e.g., sub-100 Hz frequencies) and/or frequencies that are significantly above 100 Hz. The contextual component can represent movements that the user **106** performs in order to speak, such as jaw and/or tongue movements. Additionally or alternatively, the contextual component can represent additional movements that the user **106** performs while speaking, such as blinking, rolling their eyes, or shaking their head. The contextual component can also include information regarding the user **106**'s respiration rate and/or heart rate.

Speech Recognition

[0088] FIG. 7 illustrates an example implementation of the speech-recognition module **420** for performing speech recognition **112** using active acoustic sensing. In the depicted configuration, the speech-recognition module **420** optionally includes a denoising filter **702**. The speech-recognition module **420** also includes at least one input formatter **704**, at least one machine-learned model **706**, and at least one decoder **708**. In this example, the speech-recognition module **420** is implemented using the machine-learned model **706**. Other examples are also possible in which the speech-recognition module **420** uses other signal processing and/or data analysis techniques.

[0089] The denoising filter **702** performs aspects of external-noise-source filtering and/or internal-noise-source filtering. External-noise-source filtering involves attenuating interference that is present within an external environment (e.g., a noisy environment). This interference can include any of the noise sources described with respect to FIG. 2. During operation, this noise can be unintentionally modulated onto or mixed with the ultrasound receive signal **504**. This interference can occur due to non-linearities in the microphone **410**, intermodulation distortion, harmonics, a mixing operation performed by the hearable **102**, or some other component and/or operation of the hearable **102**.

[0090] Internal-noise-source filtering involves attenuating interference within the ultrasound receive signal **504** to improve sensitivity and accuracy for audioplethysmography **110**. This interference can be caused by the hearable **102** performing other operations (e.g., rendering audio content, performing active-noise cancellation, or operating in accordance with a transparency

mode). The performance improvement associated with internal-noise-source filtering enables audioplethysmography **110** to be performed while the hearable performs these other operations. Furthermore, it improves the ability of audioplethysmography **110** to be used for speech recognition **112**.

[0091] Generally speaking, the denoising filter **702** attenuates noise that is present within the pre-processed signal **610** such that a voice component associated with the user **106**'s speech is enhanced (or amplified relative to a noise level). In this way, the denoising filter **702** can increase sensitivity for performing speech recognition **112**. Generally speaking, the voice component is superimposed onto the ultrasound receive signal **504** and is correlated with the audio receive signal **616**. The voice component is uncorrelated with the noise component.

[0092] The denoising filter **702** can be implemented using at least one internal-noise-source filter **710** (INS filter **710**) and/or at least one external-noise-source filter **712** (ENS filter **712**). The internal-noise-source filter **710** performs internal-noise-source filtering and can be implemented using at least two adaptive filters. The adaptive filters can apply various adaptive filtering techniques, including techniques based on least mean squares (LMS) or recursive least squares (RLS). A first adaptive filter performs adaptive filtering using a noise signal (e.g., a self-generated noise signal) as a noise reference to attenuate an internal-noise-source component within the pre-processed signal **610**. Likewise, a second adaptive filter performs adaptive filtering using the noise signal as a noise reference to attenuate the internal-noise-source component within the audio receive signal **616**. In this example, the noise signal is not correlated with the desired voice component within the pre-processed signal **610** and/or the audio receive signal **616**. As such, the noise signal can be used as a reference signal to attenuate the internal noise component within the pre-processed signal **610** and/or the audio receive signal **616** using adaptive filtering techniques. The noise signal can represent audio content that is rendered by the speaker **408** (or another speaker). Example types of audio content can include music, a human voice, or some other type of sound. Additionally or alternatively, the noise signal can represent an anti-noise signal that is generated by the active-noise-cancellation circuit **424** for active noise cancellation and/or a transparency-mode signal that amplifies sound from an external environment in accordance with a transparency mode of the hearable **102**.

[0093] The external-noise-source filter **712** performs external-noise-source filtering and can be implemented using at least one adaptive filter and/or at least one blind-source separator. The adaptive filter performs adaptive filtering using the audio receive signal **616** as a reference to filter a noise component from the pre-processed signal **610**. Similarly, the blind-source separator performs blind-source separation (BSS) using the audio receive signal **616** as a reference to filter the noise component from the pre-processed signal **610**. Explained another way, adaptive filtering and/or blind-source separation utilize the audio receive signal **616** to separate a voice component from the noise component within the pre-processed signal **610**. To perform adaptive filtering or blind-source separation, the pre-processed signal **610** represents a primary reference (e.g., the primary channel or the signal to be filtered) and the audio receive signal **616** represents a secondary or a noise reference (e.g., the reference channel).

[0094] The input formatter **704** appropriately formats one or more signals that are passed as inputs to the machine-learned model **706**. The input formatter **704** can be implemented using at least one spectrogram generator **714**. In some implementations, the input formatter **704** also includes a stacker **716**. The spectrogram generator **714** and the stacker **716** are further explained below.

[0095] The machine-learned model **706** is implemented using one or more neural networks. A neural network includes a group of connected nodes (e.g., neurons or perceptrons), which are organized into one or more layers. As an example, the machine-learned model **706** includes a deep neural network, which includes an input layer, an output layer, and one or more hidden layers positioned between the input layer and the output layers. The nodes of the deep neural network can be partially-connected or fully-connected between the layers.

[0096] In some implementations, the neural network is a recurrent neural network (e.g., a long short-term memory (LSTM) neural network) with connections between nodes forming a cycle to retain information from a previous portion of an input data sequence for a subsequent portion of the input data sequence. In other cases, the neural network is a feed-forward neural network in which the connections between the nodes do not form a cycle. Additionally or alternatively, the machine-learned model **706** includes another type of neural network, such as a convolutional neural network. The machine-learned model **706** can also include one or more types of regression models, such as a single linear regression model, multiple linear regression models, logistic regression models, step-wise regression models, multi-variate adaptive regression splines, locally estimated scatterplot smoothing models, and so forth. In example implementations, the machine-learned model **706** can be implemented using a single-channel-input machine-learned model **718** or a multi-channel-input machine-learned model **720**, as further described with respect to FIGS. **8** and **9**.

[0097] In general, the machine-learned model **706** is trained using supervised learning to extract features from at least a version of the ultrasound receive signal **504** such that the features can be used to recognize at least one spoken phrase **212**. The supervised learning can use simulated (e.g., synthetic) data or measured (e.g., real) data for training purposes. Features that are extracted by the machine-learned model **706** are passed to the decoder **708**.

[0098] The decoder **708** uses these extracted features to generate the recognized phrase **612**. The decoder **708** can utilize an acoustic model, a dictionary, and/or a language model to generate the recognized phrase **612**. In some cases, the recognized phrase **612** can be a transcript of the recognized phrase. In other cases, the recognized phrase **612** can be some type of digital information or signal that represents the phrase **212**.

[0099] Consider an example implementation that includes the denoising filter **702**. During an operation of the hearable **102**, the denoising filter **702** generates at least one filtered signal **722** by filtering the pre-processed signal **610** and/or the audio receive signal **616** using the techniques described above with respect to internal-noise-source filtering and/or external-noise-source filtering. The input formatter **704** generates at least one spectrogram **724** based on the filtered signal **722**. Generally speaking, the input formatter **704** generates an output that represents an input signal in a frequency vs time domain. In an example implementation, the spectrogram generator **714** applies a Fourier transform to generate a spectrogram **724** of the filtered signal **722**. A spectrogram **724** includes amplitudes associated with each frequency that is present in the filtered signal **722** across multiple time segments. The machine-learned model **706** generates a feature vector **726** based on the spectrogram **724**. The decoder **708** generates the recognized phrase **612** based on the feature vector **726**.

[0100] Other implementations of the speech-recognition module **420** can pass the audio receive signal **616** (or a filtered version thereof) to the input formatter **704**. The one or more spectrograms **724** that are provided to the machine-learned model **706** can include a spectrogram **724** of the audio receive signal **616**. Example implementations of a speech-recognition module **420** that utilizes both the pre-processed signal **610** and the audio receive signal **616** are further described with respect to FIGS. **8** and **9**.

[0101] FIG. **8** illustrates a first example implementation of the speech-recognition module **420**. In the depicted configuration, the speech-recognition module **420** includes the spectrogram generator **714**, the stacker **716**, and the single-channel-input machine-learned model **718**. The single-channel-input machine-learned model **718** can be implemented using a convolutional neural network or a single-channel transformer with a convolutional layer.

[0102] During operation, the spectrogram generator **714** generates spectrograms **802** and **804** of the pre-processed signal **610** and the audio receive signal **616**, respectively. The spectrograms **802** and **804** include spectral features **806** (e.g., amplitudes associated with different frequencies) across multiple frames **808**. The stacker **716** combines the spectrograms **802** and **804** together to generate

a stacked spectrogram **810**. In this example, the spectrogram **802** is stacked “on top of” the spectrogram **804**. Generally speaking, the stacker **716** concatenates the spectrogram **802** and the spectrogram **804** in some manner to generate the stacked spectrogram **810**. In this sense, the spectrogram **804** augments the feature size of the information provided to the single-channel-input machine-learned model **718**.

[0103] FIG. **9** illustrates a second example implementation of the speech-recognition module **420**. In the depicted configuration, the speech-recognition module **420** includes the spectrogram generator **714** and the multi-channel-input machine-learned model **720**. The multi-channel-input machine-learned model **720** can be implemented using a multiple-input convolutional neural network or a multi-channel transformer having multiple separate convolutional layers. In this example, the spectrograms **802** and **804** are provided as independent or separate inputs to the multi-channel-input machine-learned model **720**. This enables the multi-channel-input machine-learned model **720** to compute the self and cross channel correlation, which enables it to further exploit the similarities and differences between the channels for speech recognition **112**.

[0104] The example implementations shown in FIGS. **8** and **9** can support ultrasound-fusion speech recognition. Also, the implementations shown in FIGS. **8** and **9** can perform ultrasound-based speech recognition during situations in which the audio receive signal **616** does not include the voice component because the phrase **212** is spoken silently by the user **106**. Other implementations are also possible. If the speech-recognition module **420** is designed to support ultrasound-based speech recognition without supporting ultrasound-fusion speech recognition, the speech-recognition module **420** can be implemented using a spectrogram generator **714** and the single-channel-input ML model **718**. In this case, the speech-recognition module **420** can be implemented without the stacker **716**.

[0105] FIGS. **10-12** illustrate the impact of a spoken phrase **212** on an ultrasound receive signal **504**. More specifically, FIGS. **10-12** depict example amplitudes and phases of pre-processed signals **610** generated by different hearables **102-1** and **102-2**. As shown below, the pressure wave caused by a spoken phrase **212** can significantly impact the amplitude and/or the phase of the pre-processed signals **610**. In some instances, the change in the amplitude and/or the phase can be relative to a previous state or relative to a previous trend in the amplitude and/or the phase. The previous state can refer to values of the amplitude and/or the phase during which the user **106** does not speak.

[0106] In general, the term “significantly” can mean that the values of the amplitude and/or the phase can change by 20% or more relative to a previous value (e.g., relative to an average of a set of previous values). Additionally or alternatively, a slope of the amplitude and/or the phase can vary significantly. Sometimes the slope of the amplitude and/or the phase can change signs (e.g., from a positive slope to a negative slope, or vice versa). A magnitude of the slope of the amplitude and/or the phase can sometimes change by approximately 10% or more.

[0107] In some implementations, the speech-recognition module **420** can detect and recognize the phrase **212** based on the amplitude of the pre-processed signal **610** provided by the hearable **102-1**, the phase of the pre-processed signal **610** provided by the hearable **102-1**, the amplitude of the pre-processed signal **610** provided by the hearable **102-2**, the phase of the pre-processed signal **610** provided by the hearable **102-2**, or some combination thereof. Generally speaking, processing a larger quantity of signals and/or tones **604** that are sensitive to the pressure wave caused by the spoken phrase **212** provides more information to the speech-recognition module **420**. This can make it easier for the speech-recognition module **420** to accurately perform speech recognition **112**.

[0108] Graphs **1000-1** and **1000-2** in FIG. **10** depict amplitudes **1002** and phases **1004** of pre-processed signals **610** that are respectively generated by the hearables **102-1** and **102-2**. Time is depicted along the horizontal axes of the graphs **1000-1** and **1000-2**.

[0109] During the time interval indicated at **1006**, the user **106** speaks a first phrase **212-1** (e.g., audibly speaks, silently speaks, hums, whistles, sings, or makes other utterances). This causes the

amplitude **1002** and/or the phase **1004** of the ultrasound receive signal **504** to change significantly relative to a previous state. With audioplethysmography **110**, the speech-recognition module **420** can detect and recognize the phrase **212** based on the change in the amplitude **1002** and/or phase **1004** of the pre-processed signals **610** provided by the hearable **102-1** and/or the hearable **102-2**. [0110] Graphs **1100-1** and **1100-2** in FIG. **11** depict amplitudes **1002** and phases **1004** of pre-processed signals **610** that are respectively generated by the hearables **102-1** and **102-2**. Time is depicted along the horizontal axes of the graphs **1100-1** and **1100-2**.

[0111] During the time interval indicated at **1106**, the user **106** speaks a second phrase **212-2** (e.g., audibly speaks, silently speaks, hums, whistles, sings, or makes other utterances). This causes the amplitude **1002** and/or the phase **1004** of the ultrasound receive signal **504** to change significantly relative to a previous state. With audioplethysmography **110**, the speech-recognition module **420** can detect and recognize the phrase **212** based on the change in the amplitude **1002** and/or phase **1004** of the pre-processed signals **610** provided by the hearable **102-1** and/or the hearable **102-2**.

[0112] Graphs **1200-1** and **1200-2** in FIG. **12** depict amplitudes **1002** and phases **1004** of pre-processed signals **610** that are respectively generated by the hearables **102-1** and **102-2**. Time is depicted along the horizontal axes of the graphs **1200-1** and **1200-2**.

[0113] During the time interval indicated at **1206**, the user **106** speaks a third phrase **212-3** (e.g., audibly speaks, silently speaks, hums, whistles, sings, or makes other utterances). This causes the amplitude **1002** and/or the phase **1004** of the ultrasound receive signal **504** to change significantly relative to a previous state. With audioplethysmography **110**, the speech-recognition module **420** can detect and recognize the phrase **212** based on the change in the amplitude **1002** and/or phase **1004** of the pre-processed signals **610** provided by the hearable **102-1** and/or the hearable **102-2**.

[0114] As seen in FIGS. **10-12**, different phrases **212-1** to **212-3** can cause the amplitude **1002** and/or phase **1004** of the pre-processed signal **610** to vary in different ways. This further enables the speech-recognition module **420** to recognize more than one phrase **212**.

[0115] Aspects of speech recognition **112** can be performed using one hearable **102** (e.g., the hearable **102-1** or **102-2**) or multiple hearables **102** (e.g., the hearables **102-1** and **102-2**). With multiple hearables **102** performing speech recognition **112**, the computing device **104** can have higher confidence that the user **106**'s spoken phrase **212** is recognized. In general, the hearable **102** can recognize speech by analyzing changes in the amplitude **1002** of the ultrasound receive signal **504**, changes in the phase **1004** of the ultrasound receive signal **504**, or changes in both the amplitude **1002** and phase **1004** of the ultrasound receive signal **504**.

Example Methods

[0116] FIGS. **13** and **14** depict example methods **1300** and **1400** for implementing aspects of speech recognition **112** using active acoustic sensing. Methods **1300** and **1400** are shown as sets of operations (or acts) performed but not necessarily limited to the order or combinations in which the operations are shown herein. Further, any of one or more of the operations may be repeated, combined, reorganized, or linked to provide a wide array of additional and/or alternate methods. In portions of the following discussion, reference may be made to the environments **100** and **200** of FIGS. **1** and **2**, and entities detailed in FIGS. **3** and **4**, reference to which is made for example only. The techniques are not limited to performance by one entity or multiple entities operating on one device.

[0117] At **1302**, an ultrasound transmit signal is transmitted during a first time period. The ultrasound transmit signal propagates within at least a portion of an ear canal of a person. For example, the transducer **406** (or speaker **408**) of the hearable **102** transmits the ultrasound transmit signal **502**. The ultrasound transmit signal **502** propagates within at least a portion of the ear canal **114** of a person (e.g., the user **106**), as described with respect to FIG. **5**.

[0118] At **1304**, an ultrasound receive signal is received. The ultrasound receive signal represents a version of the ultrasound transmit signal with one or more waveform characteristics modified based on the propagation within the ear canal and based on the person speaking a phrase during at least a

portion of the first time period. For example, the transducer **406** (or the microphone **410**) of the hearable **102** receives the ultrasound receive signal **504**. The ultrasound receive signal **504** represents a version of the ultrasound transmit signal **502** with one or more waveform characteristics modified based on the propagation within the ear canal **114** and based on the user **106** speaking a phrase **212** during at least a portion of the first time period. The user **106** can speak the phrase **212** by audibly speaking, silently speaking, humming, singing, whispering, shouting, and so forth.

[0119] The hearable **102** that receives the ultrasound receive signal **504** can be a same hearable **102** that transmitted the ultrasound transmit signal **502** (e.g., the hearable **102-1** or **102-2** in FIG. 5), or another hearable **102** that did not transmit the ultrasound transmit signal **502** (e.g., the hearable **102-2** in FIG. 5). Example waveform characteristics include amplitude, phase, and/or frequency. In some implementations, a feedback microphone of an active-noise-cancellation circuit **424** can receive the ultrasound receive signal **504**.

[0120] At **1306**, the spoken phrase is recognized based on the ultrasound receive signal. For example, the hearable **102** uses the speech-recognition module **420** to analyze the one or more modified characteristics of the ultrasound receive signal **504** and recognize the spoken phrase **212**. The hearable **102** can generate a recognized phrase **612**, which can be used to control an operation of the hearable **102** and/or an operation of the computing device **104**. The recognized phrase **612** represents and/or identifies the phrase **212** that is spoken by the user **106**. In some implementations, the spoken phrase is recognized using a combination of the ultrasound receive signal **504** and an audio receive signal **616** that is provided by passive audio sensing.

[0121] At **1402** in FIG. 14, active acoustic sensing is performed to detect a pressure wave that propagates within an ear canal of a person and is associated with a phrase that is spoken by the person. For example, the hearable **102** performs active acoustic sensing to detect a pressure wave that propagates within an ear canal **114** of a user **106** and is associated with a phrase **212** that is spoken by the user **106**. To perform active acoustic sensing, the hearable **102** transmits and receives an ultrasound signal (e.g., the ultrasound transmit signal **502** and the ultrasound receive signal **504**). The ultrasound receive signal **504** includes a voice component, which enables audioplethysmography **110** to perform speech recognition **112**.

[0122] At **1404**, speech recognition is performed based on the active acoustic sensing. For example, the hearable **102** performs speech recognition **112** based on the active acoustic sensing. More specifically, the hearable **102** analyzes a spectrogram of the ultrasound receive signal **504** to recognize the phrase **212**. In some implementations, the hearable **102** can perform speech recognition **112** using a combination of the ultrasound receive signal **504** and the received audio signal **614**.

[0123] At **1406**, a signal that represents the phrase spoken by the person is generated. For example, the speech-recognition module **420** generates the recognized phrase **612**, which represents the phrase **212** that is spoken by the user **106**. In some implementations, this recognized phrase **612** controls an operation of at least one of the hearable **102** or the computing device **104** that is coupled to the hearable **102** is generated.

Example Computing System

[0124] FIG. 15 illustrates various components of an example computing system **1500** that can be implemented as any type of client, server, and/or computing device as described with reference to the previous FIGS. 3 and 4 to implement aspects of active acoustic sensing using a hearable **102**.

[0125] The computing system **1500** includes communication devices **1502** that enable wired and/or wireless communication of device data **1504** (e.g., received data, data that is being received, data scheduled for broadcast, or data packets of the data). The communication devices **1502** or the computing system **1500** can include one or more hearables **102**. The device data **1504** or other device content can include configuration settings of the device, media content stored on the device, and/or information associated with a user of the device. Media content stored on the computing

system **1500** can include any type of audio, video, and/or image data. The computing system **1500** includes one or more data inputs **1506** via which any type of data, media content, and/or inputs can be received, such as human utterances, user-selectable inputs (explicit or implicit), messages, music, television media content, recorded video content, and any other type of audio, video, and/or image data received from any content and/or data source.

[0126] The computing system **1500** also includes communication interfaces **1508**, which can be implemented as any one or more of a serial and/or parallel interface, a wireless interface, any type of network interface, a modem, and as any other type of communication interface. The communication interfaces **1508** provide a connection and/or communication links between the computing system **1500** and a communication network by which other electronic, computing, and communication devices communicate data with the computing system **1500**.

[0127] The computing system **1500** includes one or more processors **1510** (e.g., any of microprocessors, controllers, and the like), which process various computer-executable instructions to control the operation of the computing system **1500**. Alternatively or in addition, the computing system **1500** can be implemented with any one or combination of hardware, firmware, or fixed logic circuitry that is implemented in connection with processing and control circuits which are generally identified at **1512**. Although not shown, the computing system **1500** can include a system bus or data transfer system that couples the various components within the device. A system bus can include any one or combination of different bus structures, such as a memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or local bus that utilizes any of a variety of bus architectures.

[0128] The computing system **1500** also includes a computer-readable medium **1514**, such as one or more memory devices that enable persistent and/or non-transitory data storage (i.e., in contrast to mere signal transmission), examples of which include random access memory (RAM), non-volatile memory (e.g., any one or more of a read-only memory (ROM), flash memory, EPROM, EEPROM, etc.), and a disk storage device. The disk storage device may be implemented as any type of magnetic or optical storage device, such as a hard disk drive, a recordable and/or rewriteable compact disc (CD), any type of a digital versatile disc (DVD), and the like. The computing system **1500** can also include a mass storage medium device (storage medium) **1516**.

[0129] The computer-readable medium **1514** provides data storage mechanisms to store the device data **1504**, as well as various device applications **1518** and any other types of information and/or data related to operational aspects of the computing system **1500**. For example, an operating system **1520** can be maintained as a computer application with the computer-readable medium **1514** and executed on the processors **1510**. The device applications **1518** may include a device manager, such as any form of a control application, software application, signal-processing and control module, code that is native to a particular device, a hardware abstraction layer for a particular device, and so on.

[0130] The device applications **1518** also include any system components, engines, or managers to implement audioplethysmography **110** for speech recognition **112**. In this example, the device applications **1518** include the pre-processing module **418**, the speech-recognition module **420** (SR module **420**), and optionally the calibration module **422**. Although not explicitly shown, the device applications **1518** can also include the application **306**, the voice user interface **202**, and/or the voice authenticator **308**.

[0131] Throughout this disclosure, examples are described where a computing system **1500** (e.g., the hearable **102**, the computing device **104**, a client device, a server device, a computer, or another type of computing system) may analyze information (e.g., various audible and/or ultrasound signals) associated with a user **106**, for example, the phrase **212** mentioned with respect to FIG. 2. Further to the descriptions above, a user **106** may be provided with controls allowing the user **106** to make an election as to both if and when systems, programs, and/or features described herein may enable collection of information (e.g., information about a user **106**'s social network, social actions,

social activities, profession, a user **106**'s preferences, a user **106**'s current location), and if the user **106** is sent content or communications from a server. The computing system **1500** can be configured to only use the information after the computing system **1500** receives explicit permission from the user **106** to use the data. For example, in situations where the hearable **102** analyzes signals to recognize the user **106**'s speech, individual users **106** may be provided with an opportunity to provide input to control whether programs or features of the computing system **1500** can collect and make use of the data. Further, individual users **106** may have constant control over what programs can or cannot do with the information.

[0132] In addition, information collected may be pre-treated in one or more ways before it is transferred, stored, or otherwise used, so that personally-identifiable information is removed. For example, before the computing system **1500** shares data with another device, a user **106**'s identity may be treated so that no personally identifiable information can be determined for the user **106**. Thus, the user **106** may have control over whether information is collected about the user **106** and the user **106**'s device, and how such information, if collected, may be used by the computing system **1500** and/or a remote computing system.

CONCLUSION

[0133] Although techniques using, and apparatuses including, performing speech recognition using active acoustic sensing have been described in language specific to features and/or methods, it is to be understood that the subject of the appended examples is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed as example implementations of performing speech recognition using active acoustic sensing.

[0134] Some examples are described below:

[0135] Example 1: A method comprising: [0136] transmitting, during a first time period, an ultrasound transmit signal that propagates within at least a portion of an ear canal of a person; [0137] receiving, during the first time period, an ultrasound receive signal, the ultrasound receive signal representing a version of the ultrasound transmit signal with one or more characteristics modified based on the propagation within the ear canal and based on the person speaking a phrase during at least a portion of the first time period; and [0138] recognizing the spoken phrase based on the ultrasound receive signal.

[0139] Example 2: The method of example 1, further comprising at least one of the following:

[0140] generating a control signal that controls an operation of a device based on the spoken phrase; or [0141] generating text based on the spoken phrase.

[0142] Example 3: The method of example 2, wherein: [0143] the device comprises a hearable;

[0144] the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal using the hearable; and [0145] the receiving of the ultrasound receive signal comprises receiving the ultrasound receive signal using the hearable.

[0146] Example 4: The method of example 2, wherein: [0147] the device comprises a computing device that is coupled to a hearable; [0148] the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal using the hearable; and [0149] the receiving of the ultrasound receive signal comprises receiving the ultrasound receive signal using the hearable.

[0150] Example 5: The method of any previous example, further comprising: [0151] receiving an audio signal comprising the spoken phrase; and [0152] wherein the recognizing of the spoken phrase comprises recognizing the spoken phrase based on the ultrasound receive signal and the audio signal.

[0153] Example 6: The method of example 5, wherein the recognizing of the spoken phrase further comprises: [0154] generating a first spectrogram of a signal derived from the ultrasound receive signal; [0155] generating a second spectrogram of the audio signal; and [0156] generating a feature vector using a machine-learned model by providing the machine-learned model the first spectrogram and the second spectrogram; and [0157] recognizing the spoken phrase based on the

feature vector.

[0158] Example 7: The method of example 6, further comprising: [0159] generating a stacked spectrogram comprising a combination of the first spectrogram and the second spectrogram, [0160] wherein the generating of the feature vector comprises generating the feature vector using the machine-learned model by providing the machine-learned model the stacked spectrogram as an input.

[0161] Example 8: The method of example 7, wherein the machine-learned model comprises: [0162] a convolutional neural network; or [0163] a single-channel transformer having a convolutional layer.

[0164] Example 9: The method of example 6, wherein: [0165] the generating of the feature vector comprises generating the feature vector using the machine-learned model by providing the first spectrogram and the second spectrogram as separate inputs to the machine-learned model; and [0166] the machine-learned model comprises: [0167] a multiple-input convolutional neural network; or [0168] a multi-channel transformer comprising separate convolutional layers.

[0169] Example 10: The method of any previous example, wherein the spoken phrase is silently spoken by the person during at least the portion of the first time period.

[0170] Example 11: The method of any previous example, further comprising: [0171] transmitting, prior to the transmitting of the ultrasound transmit signal, another ultrasound transmit signal that propagates within at least the portion of the ear canal of the person, the other ultrasound transmit signal having multiple tones; [0172] receiving, prior to the transmitting of the ultrasound transmit signal, another ultrasound receive signal, the other ultrasound receive signal representing a version of the other ultrasound transmit signal with one or more characteristics modified due to the propagation within the ear canal; [0173] generating quality metrics that respectively correspond to the multiple tones, the quality metrics based on amplitudes and/or phases of the multiple tones; and [0174] selecting at least two tones from the multiple tones based on the quality metrics corresponding to the at least two tones being greater than a threshold, [0175] wherein the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal having the at least two tones.

[0176] Example 12: The method of example 11, wherein the transmitting of the ultrasound transmit signal comprises at least one of the following: [0177] transmitting the ultrasound transmit signal such that the ultrasound transmit signal has a higher amplitude at the at least two tones compared to an amplitude of the other ultrasound transmit signal at the multiple tones; or [0178] transmitting the ultrasound signal such that a duration of ultrasound transmit signal at each of the at least two tones is longer compared to a duration of the other ultrasound transmit signal at each of the multiple tones.

[0179] Example 13: The method of any previous example, wherein the recognizing of the spoken phrase comprises recognizing the spoken phrase using the ultrasound receive signal and without using one or more of the following: [0180] an audio signal that includes the spoken phrase and is captured using passive audio sensing; or [0181] another signal obtained from another sensor of the hearable that is different from an ultrasound sensor.

[0182] Example 14: The method of any previous example, further comprising: [0183] rendering audible content during the first time period, the rendering causing an audible signal to propagate within at least a portion of the ear canal of the person.

[0184] Example 15: The method of any previous example, wherein: [0185] the rendering of the audible content comprises transmitting, during the first time period, audible signal that propagates within at least a portion of the ear canal of the person; [0186] the ultrasound receive signal comprises an internal noise component caused by interference generated by the rendering of the audible signal; [0187] the method further comprises generating a denoised signal by filtering the internal noise component within the ultrasound receive signal based on a version of the audible signal; and [0188] the recognizing of the spoken phrase comprises recognizing the spoken phrase

based on the denoised signal.

[0189] Example 16: A non-transitory computer-readable storage medium comprising instructions that, responsive to execution by a processor, cause a hearable to perform any one of the methods of examples 1 to 15.

[0190] Example 17: A device comprising: [0191] at least one transducer; and [0192] at least one processor, the device configured to perform, using the at least one transducer and the at least one processor, any one of the methods of examples 1 to 15.

[0193] Example 18: The device of example 17, further comprising: [0194] a speaker; and [0195] an active-noise-cancellation circuit comprising a feedback microphone, [0196] wherein the at least one transducer comprises the speaker and the feedback microphone.

[0197] Example 19: The device of example 17, wherein: [0198] the at least one transducer comprises a speaker and a microphone; [0199] the speaker is configured to be positioned proximate to a first ear of a person; and [0200] the microphone is configured to be positioned proximate to a second ear of the person.

[0201] Example 20: The device of any one of examples 17 to 19, wherein the device comprises: at least one earbud.

Claims

1. A method comprising: transmitting, during a first time period, an ultrasound transmit signal that propagates within at least a portion of an ear canal of a person; receiving, during the first time period, an ultrasound receive signal, the ultrasound receive signal representing a version of the ultrasound transmit signal with one or more characteristics modified based on the propagation within the ear canal and based on the person speaking a phrase during at least a portion of the first time period; and recognizing the spoken phrase based on the ultrasound receive signal.
2. The method of claim 1, further comprising at least one of the following: generating a control signal that controls an operation of a device based on the spoken phrase; or generating text based on the spoken phrase.
3. The method of claim 2, wherein: the device comprises a hearable; the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal using the hearable; and the receiving of the ultrasound receive signal comprises receiving the ultrasound receive signal using the hearable.
4. The method of claim 2, wherein: the device comprises a computing device that is coupled to a hearable; the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal using the hearable; and the receiving of the ultrasound receive signal comprises receiving the ultrasound receive signal using the hearable.
5. The method of claim 1, further comprising: receiving an audio signal comprising the spoken phrase; and wherein the recognizing of the spoken phrase comprises recognizing the spoken phrase based on the ultrasound receive signal and the audio signal.
6. The method of claim 5, wherein the recognizing of the spoken phrase further comprises: generating a first spectrogram of a signal derived from the ultrasound receive signal; generating a second spectrogram of the audio signal; and generating a feature vector using a machine-learned model by providing the machine-learned model the first spectrogram and the second spectrogram; and recognizing the spoken phrase based on the feature vector.
7. The method of claim 6, further comprising: generating a stacked spectrogram comprising a combination of the first spectrogram and the second spectrogram, wherein the generating of the feature vector comprises generating the feature vector using the machine-learned model by providing the machine-learned model the stacked spectrogram as an input.
8. The method of claim 7, wherein the machine-learned model comprises: a convolutional neural network; or a single-channel transformer having a convolutional layer.

9. The method of claim 6, wherein: the generating of the feature vector comprises generating the feature vector using the machine-learned model by providing the first spectrogram and the second spectrogram as separate inputs to the machine-learned model; and the machine-learned model comprises: a multiple-input convolutional neural network; or a multi-channel transformer comprising separate convolutional layers.

10. The method of claim 1, wherein the spoken phrase is silently spoken by the person during at least the portion of the first time period.

11. The method of claim 1, further comprising: transmitting, prior to the transmitting of the ultrasound transmit signal, another ultrasound transmit signal that propagates within at least the portion of the ear canal of the person, the other ultrasound transmit signal having multiple tones; receiving, prior to the transmitting of the ultrasound transmit signal, another ultrasound receive signal, the other ultrasound receive signal representing a version of the other ultrasound transmit signal with one or more characteristics modified due to the propagation within the ear canal; generating quality metrics that respectively correspond to the multiple tones, the quality metrics based on amplitudes and/or phases of the multiple tones; and selecting at least two tones from the multiple tones based on the quality metrics corresponding to the at least two tones being greater than a threshold, wherein the transmitting of the ultrasound transmit signal comprises transmitting the ultrasound transmit signal having the at least two tones.

12. The method of claim 11, wherein the transmitting of the ultrasound transmit signal comprises at least one of the following: transmitting the ultrasound transmit signal such that the ultrasound transmit signal has a higher amplitude at the at least two tones compared to an amplitude of the other ultrasound transmit signal at the multiple tones; or transmitting the ultrasound signal such that a duration of ultrasound transmit signal at each of the at least two tones is longer compared to a duration of the other ultrasound transmit signal at each of the multiple tones.

13. The method of claim 1, wherein the recognizing of the spoken phrase comprises recognizing the spoken phrase using the ultrasound receive signal and without using one or more of the following: an audio signal that includes the spoken phrase and is captured using passive audio sensing; or another signal obtained from another sensor that is different from an ultrasound sensor.

14. The method of claim 1, further comprising: rendering audible content during the first time period, the rendering causing an audible signal to propagate within at least a portion of the ear canal of the person.

15. The method of claim 14, wherein: the rendering of the audible content comprises transmitting, during the first time period, audible signal that propagates within at least a portion of the ear canal of the person; the ultrasound receive signal comprises an internal noise component caused by interference generated by the rendering of the audible signal; the method further comprises generating a denoised signal by filtering the internal noise component within the ultrasound receive signal based on a version of the audible signal; and the recognizing of the spoken phrase comprises recognizing the spoken phrase based on the denoised signal.

16. A non-transitory computer-readable storage medium comprising instructions that, responsive to execution by a processor, cause a hearable to: transmit, during a first time period, an ultrasound transmit signal that propagates within at least a portion of an ear canal of a person; receive, during the first time period, an ultrasound receive signal, the ultrasound receive signal representing a version of the ultrasound transmit signal with one or more characteristics modified based on the propagation within the ear canal and based on the person speaking a phrase during at least a portion of the first time period; and recognize the spoken phrase based on the ultrasound receive signal.

17. A device comprising: at least one transducer configured to: transmit, during a first time period, an ultrasound transmit signal that propagates within at least a portion of an ear canal of a person; and receive, during the first time period, an ultrasound receive signal, the ultrasound receive signal representing a version of the ultrasound transmit signal with one or more characteristics modified based on the propagation within the ear canal and based on the person speaking a phrase during at

least a portion of the first time period; and at least one processor configured to recognize the spoken phrase based on the ultrasound receive signal.

18. The device of claim 17, further comprising: a speaker; and an active-noise-cancellation circuit comprising a feedback microphone, wherein the at least one transducer comprises the speaker and the feedback microphone.

19. The device of claim 17, wherein: the at least one transducer comprises a speaker and a microphone; the speaker is configured to be positioned proximate to a first ear of a person; and the microphone is configured to be positioned proximate to a second ear of the person.

20. The device of claim 17, wherein the device comprises: at least one earbud.
